

Krishnaa Vadivel | Software Engineer | Scientific Computing | HPC Workflows

📞 516-853-5663 • ✉ srikrishnaa.careers@gmail.com • 🌐 krishnaa423
in sri-krishnaa-ece-phd • leetcode.com/u/krishnaa42342 | hub.docker.com/u/krishnaa42342

Profile

Software engineer and PhD researcher focused on scientific computing, distributed workflows, numerical software, and engineering tooling. Strong background in Python and C++ development for simulation, automation, data processing, visualization, and infrastructure spanning research and production-adjacent environments.

Research Experience

Yale University, Electrical Engineering

Research Assistant / PhD Researcher

2021–present

- Develop scientific software for first-principles and many-body workflows in materials research, combining Python and C++ with MPI, OpenMP, PETSc, SLEPc, and Linux-based automation.
- Build distributed and accelerator-enabled workflows for HPC environments including Frontier, Frontera, and Perlmutter, with experience spanning CUDA-, ROCm-, and container-oriented development patterns.
- Create reusable tooling, notes, and structured workflows that bridge theory, implementation, and reproducibility in computational science.
- Extend into ML-assisted and retrieval-driven automation where useful, particularly for scientific-data generation and research-workflow setup.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built CUDA-accelerated processing and custom OpenGL visualization software for large acoustic waveform datasets used in engineering diagnostics and field-tool analysis.
- Developed ASP-style services and Microsoft SQL-backed workflows for internal software and field-data management.
- Worked close to embedded and instrumentation systems, giving practical experience with software operating alongside deployed hardware and field constraints.
- Participated in broader engineering support spanning embedded-adjacent software, validation, and integration contexts.

FindLight

Photonics Technical Writer

2018–2019

- Produced technical content for optics and photonics topics, demonstrating strong technical communication across specialized engineering subject matter.

Selected Projects

DOLFINx Semiconductor and PDE Workflow

- Built Python workflows around DOLFINx and PETSc for Poisson and drift-diffusion style simulation, including theory notes, implementation scaffolding, and numerical experimentation.

Agentic Scientific Workflow Generation

- Developed retrieval-driven tooling that reads documentation and academic papers to generate structured starter inputs and workflow artifacts for computational-science tasks.

ROS 2 TurtleBot Simulation Stack

- Organized a Docker-oriented robotics/software project for navigation simulation, with software-first fallbacks and analysis artifacts.

Scientific Visualization and GPU Tooling

- Built CUDA/OpenGL-based tooling and related utilities for large numerical datasets, interactive inspection, and engineering communication.

First-Principles and Photonics Code Artifacts

- Created clean, inspectable software artifacts for plane-wave band-structure workflows and FDTD-based photonics simulation.

Education

Yale University

PhD, Electrical Engineering

2021–present

Ongoing work combining scientific computing, high-performance workflows, and computational materials research.

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Selected Coursework

Computing Foundations: Theoretical Computer Science; Probabilistic Machine Learning

Scientific Foundations: Graduate Quantum Mechanics; Solid State Physics; Many-Body Quantum Physics

Engineering Foundations: Signals and systems, embedded systems, semiconductor physics, and numerical/scientific computing practice

Technical Skills

Languages: Python, C, C++, JavaScript, Bash, PowerShell, SQL

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, NumPy, pandas, SciPy, SymPy, matplotlib

Systems: Linux command line, CMake, Docker, Docker Compose, Kubernetes, gcloud

Platforms: CUDA, OpenGL, Frontier, Frontera, Perlmutter

ML / Automation: PyTorch, scikit-learn, agentic workflows, retrieval-driven automation